

$$\phi_{mn} = \frac{2}{\sqrt{\rho h L_x L_y}} \sin \frac{m\pi x}{L_x} \sin \frac{n\pi y}{L_y}$$

$$\omega_{mn} = \sqrt{\frac{D}{\rho h} \left[\left(\frac{m\pi}{L_x} \right)^2 + \left(\frac{n\pi}{L_y} \right)^2 \right]}, \quad N \text{ modes}$$